Modeling Behavioral Interaction Features in AI-Driven Learning Systems

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Contents

1 Introduction **1**
1.1 Related work 2

2 Dataset and Methodology **2**
2.1 Feature Engineering 2
2.2 Data 3
2.3 Experimental Setup 3

3 Behavioral Model **3**
3.1 Engagement (Interaction Intensity) 3
3.2 Persistence (Cognitive Effort) 4
3.3 Efficiency (Outcome Relative to Effort) 4
3.4 Friction Tolerance (pure effort) 4

4 Results and Discussion **4**
4.1 Behavioral Trends 4
4.2 Predictive Modeling of Continued Usage 6
4.3 Feature Importance Analysis 6

5 Conclusion **7**

1 Introduction

Artificial intelligence is increasingly integrated into educational environments, fundamentally reshaping how students interact with learning resources. Modern AI-driven systems provide on-demand assistance, adaptive feedback, and personalized guidance, but their effectiveness depends

not only on the availability of these tools, but on how learners engage with them. Understanding behavioral interaction patterns is therefore essential for evaluating both learning outcomes and long-term system adoption (1).

Prior research has focused on performance metrics or user satisfaction, with less attention given to the behavioral features that emerge during interaction, such as engagement intensity, persistence, and efficiency. The behavioral signals can offer deeper insight into how students balance effort, reliance on AI assistance, and perceived outcomes during study sessions(2).

This study investigates the role of behavioral metrics in predicting user satisfaction and continued usage of AI-assisted learning systems. Using a structured dataset of simulated study sessions, normalized indicators are constructed in such a way that they capture engagement, persistence, efficiency, and friction tolerance, which are later analyzed so their impact on retention and satisfaction can be approximated.

The results highlight efficiency as the dominant predictor of continued usage, suggesting that users prioritize outcome quality relative to effort over raw interaction volume. These findings contribute to the design of more effective and user-centered AI learning systems.

1.1 Related work

Research in AI-driven learning systems has shown that interaction traces such as time-on-task, activity frequency, and help-seeking behavior are strong predictors of student performance and engagement (1; 3). These approaches form the basis of learning analytics and educational data mining, where machine learning models are used to infer academic outcomes from behavioral data.

More recent studies emphasize engagement as a multidimensional construct capturing behavioral and cognitive involvement in learning activities (2). However, most existing work relies on aggregated usage statistics and does not explicitly model higher-level behavioral dynamics such as persistence under effort, efficiency of learning outcomes, or dependence on AI assistance. This limits their ability to fully represent learner interaction patterns in AI-assisted environments.

This study addresses this gap by introducing a compact behavioral framework that integrates engagement, persistence, efficiency, and friction tolerance into a unified predictive model of user behavior.

2 Dataset and Methodology

2.1 Feature Engineering

To ensure comparability across variables with different scales, raw interaction features are standardized using Z-score normalization:

$$Z(x) = \frac{x - \mu}{\sigma}$$

where μ and σ denote the mean and standard deviation of the feature, respectively.

Outcome-related variables are transformed using Min-Max scaling to map values to the interval [0,1]:

$$\text{MinMax}(x) = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

This ensures strictly positive values, which are required for ratio-based feature construction.

For stability in division operations, a small constant $\epsilon = 0.01$ is introduced to prevent division by zero. Additionally, the Engagement metric is shifted to ensure strictly positive values:

$$\text{Engagement}_{\text{shifted}} = \text{Engagement} + |\min(\text{Engagement})| + \epsilon$$

These transformations ensure numerical stability and prevent scale-related distortions in the computed behavioral metrics.

2.2 Data

The data represents a synthetic dataset that models student interaction with AI-assisted learning systems. The dataset contains 11 fields with data collected during AI-assisted study sessions. The dataset used in this study was obtained from Kaggle (4). Before any analysis was done, the data was cleaned(unnecessary fields were removed, non-numerical fields were replaced with numerical representations), behavioral features were computed(Engagement, Persistence, Efficiency, and Friction tolerance), and each feature was constructed to positively or negatively weight behavioral signals (3).

2.3 Experimental Setup

To evaluate the predictive power of behavioral features, a Random Forest classifier (5) was trained to predict continued usage of the system, defined as a binary variable indicating whether a user returns for further sessions.

The dataset was split into training and testing sets using an 80/20 stratified split to preserve class distribution. Model performance was optimized using randomized hyperparameter search with cross-validation.

The final model used 100 decision trees, with a maximum depth of 10 and Gini impurity as the splitting criterion. A fixed random seed was used to ensure reproducibility of results.

Model performance was evaluated using accuracy, ROC AUC, precision, and recall to account for class imbalance and provide a more complete assessment of predictive quality.

The full implementation of the preprocessing pipeline, feature construction, and model training procedure is provided in an accompanying Kaggle notebook to ensure reproducibility of all reported experiments.

3 Behavioral Model

The behavioral dimensions considered in this study are conceptually grounded in prior work on the ThinkR platform, where interaction-based features were shown to reflect cognitive and motivational aspects of learning behavior (6).

3.1 Engagement (Interaction Intensity)

Engagement captures the total volume of interaction, balancing both time spent and prompt frequency. The engagement feature is rewarding each prompt and each minute spent studying. The formula of engagement is the following:

$$\text{Engagement} = Z(\text{TotalPrompts}) + Z(\text{SessionLengthMin})$$

3.2 Persistence (Cognitive Effort)

Persistence measures sustained effort, it gently penalizes over-reliance on AI assistance (which requires less independent cognitive effort)and rewards the cognitive effort. The coefficient (0.3) was empirically chosen to moderate the influence of AI assistance, ensuring that assistance moderates but does not dominate the persistence signal. This reflects the assumption that moderate AI use supports learning, while excessive reliance reduces independent cognitive effort.The formula of persistence is the following:

$$\text{Persistence} = Z(\text{TotalPrompts}) - (0.3 \times Z(\text{AI_AssistanceLevel}))$$

3.3 Efficiency (Outcome Relative to Effort)

Efficiency computes the successful output realative to engagement, it is modeled as outcome quality relative to interaction cost. The formula of efficiency is the following:

$$\text{Efficiency} = \frac{MinMax(\text{FinalOutcome}) \times MinMax(\text{SatisfactionRating})}{\text{Engagement}_{\text{shifted}} + \epsilon}$$

3.4 Friction Tolerance (pure effort)

The Friction Tolerance metric captures cognitive effort, by penalizing reliance on AI assistance. Higher values indicate users who engage actively while maintaining lower dependence on automated support. While less expressive than composite features, it serves as a baseline indicator of autonomy in interaction behavior.The formula of friction tolerance is the following:

$$\text{FrictionTolerance} = \text{TotalPrompts} - \text{AI_AssistanceLevel}$$

Together, these metrics form a structured behavioral representation of user interaction, balancing effort, outcome, and assistance. This feature space enables the subsequent analysis of their relative importance in predicting satisfaction and continued usage.

4 Results and Discussion

Following the extraction and normalization of behavioral features, their relationships with learning outcomes, user satisfaction, and continued usage were analyzed. Both exploratory analysis and predictive modeling were employed to identify the most influential behavioral signals.

4.1 Behavioral Trends

Engagement. Engagement exhibits a strong positive relationship with both learning outcomes and user satisfaction. Higher engagement scores, derived from increased interaction frequency and session duration, consistently correspond to improved final outcomes. This trend suggests that active and sustained interaction with AI-assisted systems contributes positively to task completion and

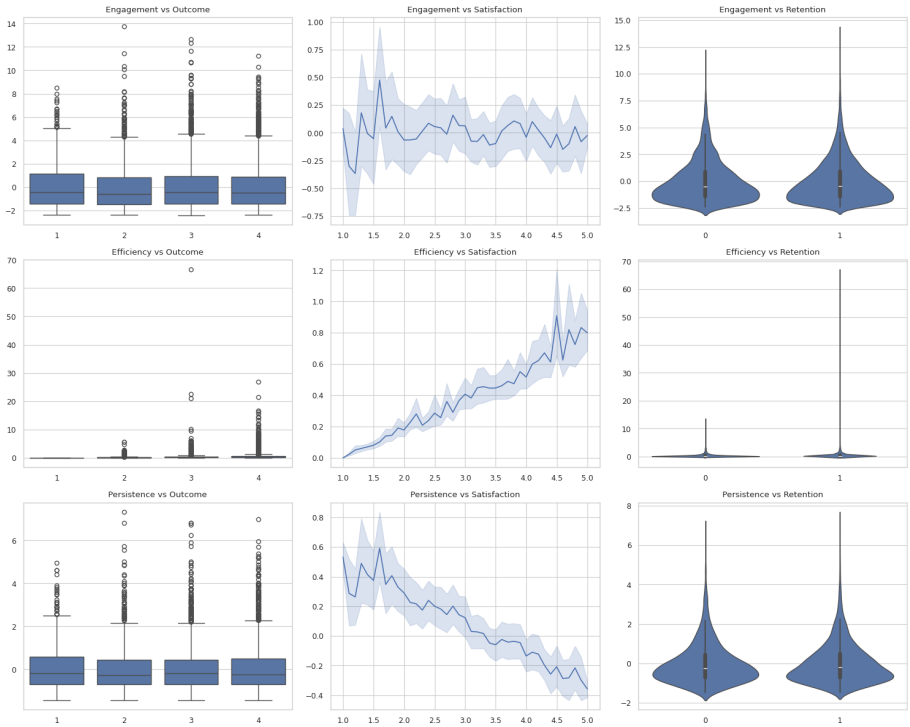

Figure 1: Behavioral feature relationships across outcome, satisfaction, and retention dimensions. Rows represent behavioral metrics (Engagement, Efficiency, Persistence), and columns correspond to Final Outcome, Satisfaction Rating, and Continued Usage. Efficiency emerges as the dominant predictor of retention, while Engagement shows consistent positive trends and Persistence reflects a trade-off between success and user experience.

perceived effectiveness. The presence of outliers, captured through Z-score normalization, reflects meaningful behavioral variation rather than noise, indicating distinct user interaction patterns.

Persistence. Persistence demonstrates a clear positive association with successful task completion. Users who achieve higher final outcomes tend to exhibit higher persistence scores, reflecting sustained effort and continued interaction. However, an inverse relationship is observed between persistence and satisfaction. Users reporting the highest satisfaction levels tend to require less persistence, suggesting that prolonged effort may be associated with friction or difficulty during interaction. This highlights a trade-off between effort and user experience: persistence supports success, but excessive persistence may reduce perceived usability.

Efficiency. Efficiency presents the most distinctive behavioral pattern. While lower and intermediate outcome categories show relatively low and stable efficiency values, a sharp increase is observed for sessions resulting in successful task completion. Similarly, higher satisfaction levels strongly correlate with increased efficiency. This indicates that users derive the greatest satisfaction when achieving high-quality outcomes with minimal interaction cost. Efficiency therefore captures the balance between effort and outcome, making it a critical indicator of effective system use.

4.2 Predictive Modeling of Continued Usage

The optimized model achieved an accuracy of 72.5% and an ROC AUC score of 0.65, indicating moderate predictive capability. While overall performance is satisfactory, class-level analysis reveals an imbalance in predictive sensitivity. The model achieved a recall of 0.25 for non-returning users, indicating limited detection of disengagement cases. However, precision for this class reached 0.57, suggesting that flagged at-risk users are identified with relatively high reliability.

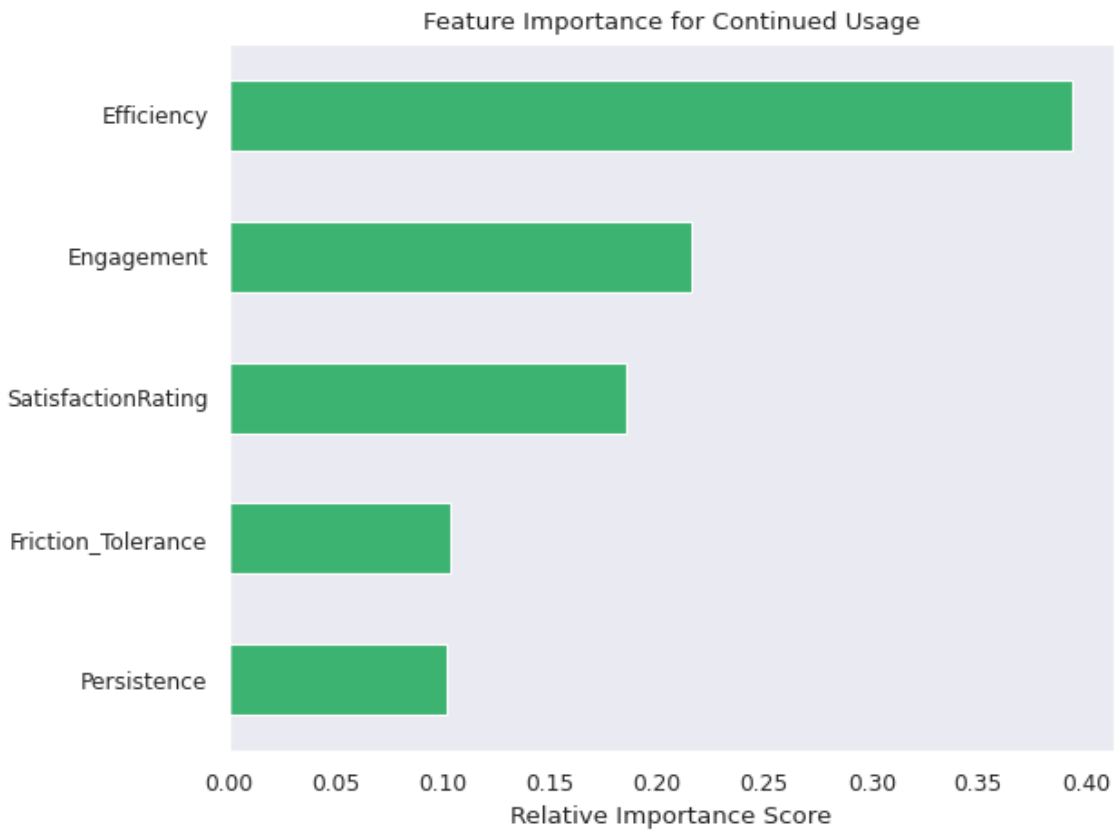

Figure 2: Feature importance scores derived from the trained Random Forest model. The plot shows the relative contribution of behavioral metrics—Engagement, Persistence, Efficiency, Friction Tolerance, and Satisfaction Rating—in predicting continued usage of the AI-assisted learning system.

4.3 Feature Importance Analysis

Feature importance analysis derived from the trained Random Forest model highlights the relative contribution of each behavioral metric:

- **Efficiency** is the dominant predictor, contributing approximately 40% of the model's decision

weight.

- **Engagement** (22%) and **Satisfaction Rating** (18%) serve as strong secondary predictors.
- **Persistence** and **Friction Tolerance** contribute the least (approximately 10% each).

These results indicate that continued usage is primarily driven by the efficiency of interaction rather than raw effort. Users are significantly more likely to return when their interactions yield high-quality outcomes relative to the time and effort invested.

5 Conclusion

This study proposed a structured framework for modeling behavioral interaction features in AI-assisted learning systems and evaluated their relationship with user satisfaction and continued usage.

The results indicate that behavioral efficiency is the most significant predictor of system retention, outperforming both engagement and effort-based metrics. This suggests that users prioritize achieving high-quality outcomes with minimal interaction cost rather than maximizing interaction intensity. Accordingly, efficiency emerges as a unifying behavioral signal that captures both performance effectiveness and perceived user experience.

These findings have direct implications for the design of AI-driven educational systems. In particular, systems should prioritize optimizing outcome efficiency and reducing unnecessary interaction overhead, rather than focusing solely on increasing engagement or interaction volume. Such design choices are more likely to support sustained user adoption and perceived usability.

The study is subject to limitations. First, the dataset is synthetic, which may not fully reflect the complexity and noise of real-world educational interactions. Second, the behavioral features are derived from predefined transformations and weighting schemes, which introduce modeling assumptions that may not capture all aspects of human learning behavior. Third, the predictive evaluation is conducted on a single dataset without external validation, limiting generalizability.

Despite these limitations, the proposed framework provides a structured approach for analyzing behavioral interaction signals in AI-assisted learning environments. Future work should validate these findings on real-world datasets and extend the analysis to temporal behavioral dynamics across extended learning trajectories.

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